Superlattices and Microstructures 107 (2017) 5–16

Contents lists available at ScienceDirect

Superlattices and Microstructures

journal homepage: www.elsevier.com/locate/superlattices

Si and Ge step-FinFETs: Work function variability, optimization and electrical parameters

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ARTICLE INFO

Article history:
Received 18 March 2017
Received in revised form 1 April 2017
Accepted 4 April 2017
Available online 5 April 2017

Keywords:
FinFET
Ge step-FinFET
Metal gate granularity
Si step-FinFET
Work function variability (WFV)

ABSTRACT

We present two different step-FinFETs under the consideration that fin material is made of either Si or Ge, named as Si step-FinFET and Ge step-FinFET. A comparative simulation study among conventional FinFET (C-FinFET), and proposed step-FinFETs is presented. It is observed that Ge step-FinFET offers better I_{on}/I_{off} ratio, with a lower intrinsic delay. Parametric analysis shows that Si step-FinFET is more resistant to subthreshold swing (SS), drain induced barrier lowering (DIBL), threshold voltage (V_T) roll off and Ge step-FinFET has higher I_{on}/I_{off} ratio, lower intrinsic delay at various channel length and oxide thickness. In presence of gate metal work function variations (WFV), a comparison of electrical parameters between C-FinFET, Si step-FinFET, and Ge step-FinFET has also been studied. We found that Si step-FinFET gives lesser variation in threshold voltage (σV_T), lesser variation in subthreshold swing (σSS), and higher variation in current ratio ($\sigma(I_{on}/I_{off})$) than C-FinFET. σV_T, σSS, and $\sigma(I_{on}/I_{off})$ are compared for Ge step-FinFET and Si step-FinFET.

1. Introduction

The requirement of the semiconductor industry to deliver ever-increasing functionality and performance in low power applications has led to continuous downscaling of MOSFET dimensions [1–3]. However, scaling in nanometre regime has reached a point where there are various short channel effects (SCEs) like drain induced barrier lowering (DIBL), threshold voltage roll-off, degradation of off current, and so forth become much more serious concerns. With the reduction of channel length, the lateral electric fields from source and drain influence the channel, and the gate losses its control over the channel. As a result, in off mode of operation the gate becomes ineffective to completely put the channel off leading to an increased off current. At the same time, scaling of oxide thickness results in high gate leakage and gate induced drain leakage (GIDL) [4,5]. Therefore, suppression of SCEs is very challenging to follow the Moore's law [6]. In order to control SCEs, novel device architectures, modification of channel/source materials, and unconventional device structures have been reported in literature. Among them, FinFET (Fin-Field Effect Transistor) [7,8] is one of the very promising devices due to its capability to control the channel from all three sides by the gate. Due to better control on the channel, FinFET has high I_{on}/I_{off} and reduced SCE. The electrical characteristics of such device are not only dependent on gate length, oxide thickness, and channel doping concentration but also a function of fin width and fin thickness. Therefore, variation of fin width and fin height influences the device performance and needs to be controlled. It has been stated that with the reduction of fin width SCE reduces but device performance become inferior [9,10].

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http://dx.doi.org/10.1016/j.spmi.2017.04.001
0749-6036/

To further improve the performance of FinFETs, many researchers have proposed various structures such as DG-FinFET [11], TG-FinFET [8] [11], GAA FinFET [11], tapered FinFET [12], Pie gate bulk FinFET [13], Omega-FinFET [14], Cylindrical FinFET [15], and Tri-gate heterojunction SOI FinFET [16]. Nowadays, metal gates are used as gate materials instead of poly-silicon gate. Metal gate has lower gate resistance which increases the on current of the device. But crystal structure of metal gate can be amorphous or polycrystalline with varying grain size and its work function (WF) is orientation dependent [17]. Due to such variations in WF at different directions, metal gate introduces a random variation of V_T within the gate area. The dependence of WF on orientations of metal grain and its impact on V_T has been presented through analytical [18] and simulation [19] study. Similarly, other researchers have reported the impact of WFV on electrical characteristics of FinFET [20–22]. Recently, Ge p-channel FinFET has been reported in presence of work function variability (WFV) [23]. In Ref. [24] systematic studies on various electrical parameters due to WFV in n step-FinFET have reported. In general, all these works exhibit appreciable results. Nevertheless, further scopes for improving subthreshold slope, DIBL effect, gate capacitance, I_{on}/I_{off} ratio etc. exists; in particular, with modifications in the device architecture.

In this paper, we have proposed a step-FinFET having its fin in the shape of step and made a comparative study of electrical parameters among C-FinFET, Si step-FinFET, and Ge step-FinFET. To optimize the dimension of Si step-FinFET and Ge step-FinFET, a parametric analysis has been carried out by varying gate length and oxide thickness. Furthermore, the impacts of WFV among C-FinFET, Si step-FinFET, and Ge step-FinFET for the different structural combinations have been investigated.

2. Device structure and simulation setup

Fig. 1(a) depicts the 3D view of the proposed n-channel step-FinFET. Fig. 1(b) and (c) show the front view of proposed FinFET and C- FinFET, respectively. The fin of C-FinFET is made of Si and the fin of step-FinFET is made of either Si or Ge. When step-FinFET is designed from C-FinFET, oxide thickness at both sides of the fin increases. Now there are two different oxide thicknesses t_{ox1}, and $t_{ox2,}$ and fin heights H_{fin1} and H_{fin2} as shown in Fig. 1(b). Oxide materials (SiO_2 and Si_3N_4) act as buried oxide and gate oxide, respectively, for all the structures. The dimensions of various parameters for different structures are listed in Table 1. The doping specifications of various region on the device are: source and drain (n-type, 10^{20} cm^{-3}), and

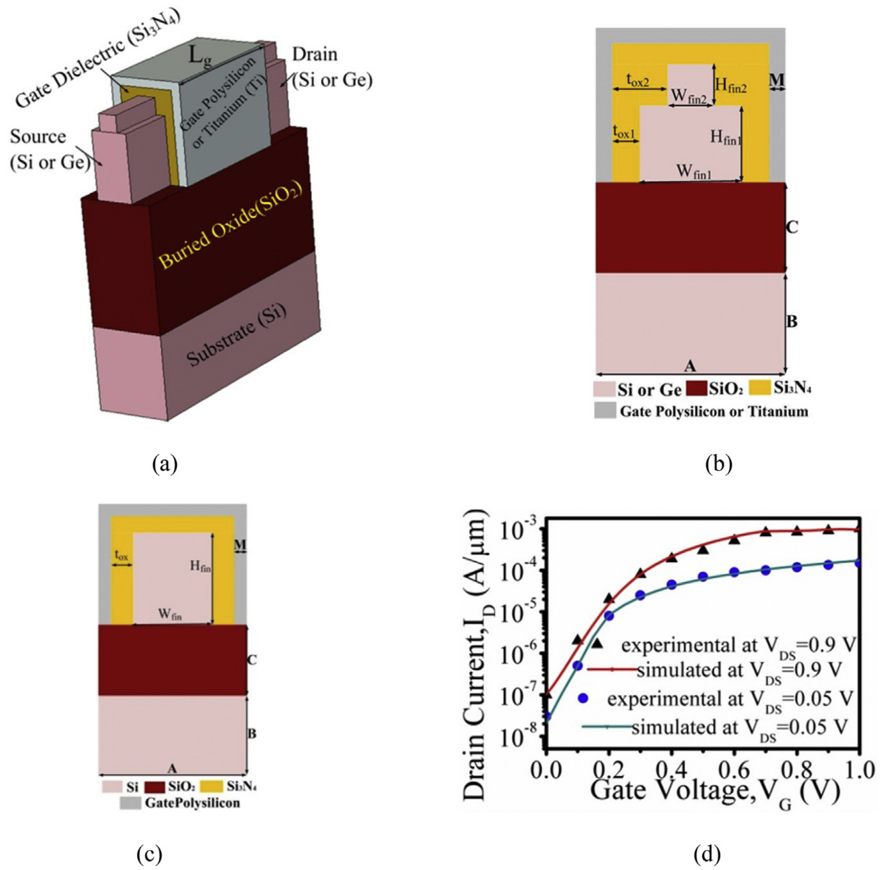


Fig. 1. (a) 3D view of proposed Si or Ge step-FinFET, (b) Front view of proposed Si or Ge step-FinFET, (c) Front view of Conventional FinFET, and (d) Calibration of simulator with experimental data [33].

Table 1
Nomenclature and values of parameters used in the simulated FinFET as shown in Fig. 1(b) and (c).

Parameter	Value (nm)
Substrate thickness (A)	8
Substrate height (B)	10
SiO_2 height (C)	15
Gate Length (L_g)	varied
Gate Metal Thickness (M)	1
Fin height at lower section of step-FinFET (H_{fin1})	8
Fin height at upper section of step-FinFET (H_{fin2})	2
Fin width at lower section of step-FinFET (W_{fin1})	varied
Fin width at upper section of step-FinFET (W_{fin2})	varied
Oxide thickness at lower section of step-FinFET (t_{ox1})	varied
Oxide thickness at upper section of step-FinFET (t_{ox2})	varied
Fin height of C-FinFET (H_{fin})	10
Fin width of C-FinFET (W_{fin})	4
Oxide thickness of C-FinFET (t_{ox})	1

channel (p-type, 10^{16} cm^{-3}), all uniformly doped. Polysilicon is used as the gate material for comparison purpose and Ti is used for WFV analysis.

Gate oxide and gate material are Si_3N_4 and gate polysilicon, respectively. Experimentally, it was observed for NMOS device that the leakage current is lower at Si_3N_4/Si interface than SiO_2/Si [25]. However, Si_3N_4 is not compatible with polysilicon gate due to two major problems: Fermi level pinning [26] and phonon scattering [27]. To tackle these problems, metal gates are used instead of polysilicon gates. Si and Ge have different material properties; so different metal gates should be used. However, WF of metal gate is direction dependent and pattern of WF (Φ_M) varies from one metal gate to another. Therefore, comparison of electrical parameters for Si and Ge fins due to WFV by using different metals is not possible. We have used Ti as gate material for WFV as it is experimentally useful [28–30] for both Ge and Si MOSFETs/FinFETs. Table 2 gives the possible grain orientations with their probabilities and WFs for Ti [31] metal gate. From this table, it is observed that Ti metal has 60% chance of having WF = 4.6 eV in <200> direction and 40% likelihood of WF = 4.4 eV in <111> direction.

The simulation results for all structures are extracted from 3D Sentaurus TCAD, Synopsys [32]. Due to degenerate source and drain regions, Fermi-Dirac Statistics is used instead of Boltzmann approximation [32]. Since high doping regions are present in the device, bandgap narrowing model and Shockley Read Hall recombination [32] models are activated. As the device is in nanoscale, doping dependent Masetti mobility model [32] is included in our simulation in order to take account of the effect of doping concentration on mobility. With these TCAD models, the simulation result is compared with experimental data in Ref. [33] as shown in Fig. 1(d). On doing calibration, some mobility parameters are modified and channel is doped with p-type (10^{15} cm^{-3}). The modified values of mobility parameters are: mumin1 = 140 cm^2/Vs, mumin2 = 25 cm^2/Vs, mu1 = 25 cm^2/Vs, $C_r = 25.7 \times 10^8$ cm^{-3}, and $\beta = 5$. A good accuracy is obtained between simulation and experimental data.

In addition to the above model, Quantum Density Gradient model [32] is also activated to take care of quantum correction (QC) effect. The effect of WFV due to Ti metal gate within the gate area is measured with the help of randomization algorithm [32] present within the simulator. In the random algorithm, we have considered nonuniform distribution with an average grain size of 5 nm as suggested in Ref. [23]. Each combination of structures has been simulated an ensemble of 200 different times and each device has unique grain pattern.

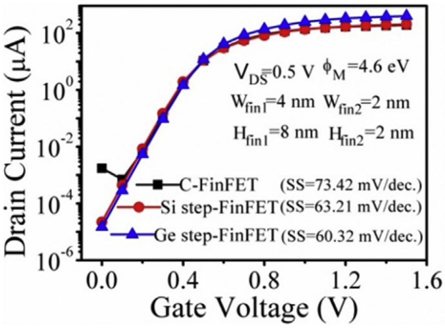
3. Results and discussion

3.1. Comparison of performance among conventional FinFET, Si step-FinFET and Ge step-FinFET

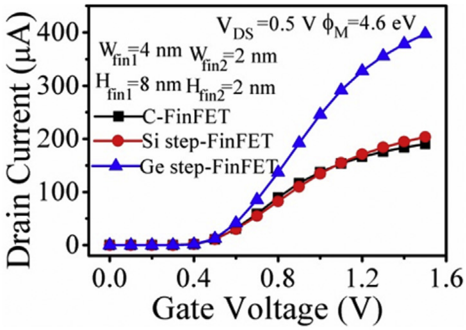
Fig. 2(a) and (b) shows the comparison of drain current among C-FinFET, Si step-FinFET, and Ge step-FinFET with $L_g = 8$ nm, $t_{ox1} = 1$ nm, and $t_{ox2} = 2$ nm in log and linear scale, respectively. It is observed that proposed step-FinFETs show high on current and low off current than C-FinFET. Drain current in linear region of step-FinFETs decreases due to decrease in fin width leading to an increase in source/drain resistance. Due to more oxide thickness, step-FinFETs provide more on current. However, Ge step-FinFET shows higher on and lesser off current than Si step-FinFET. This is due to narrow band gap as well as high mobility of Ge device.

Table 2
Possible grain orientation with their corresponding probability and WF for a Ti Metal gate.

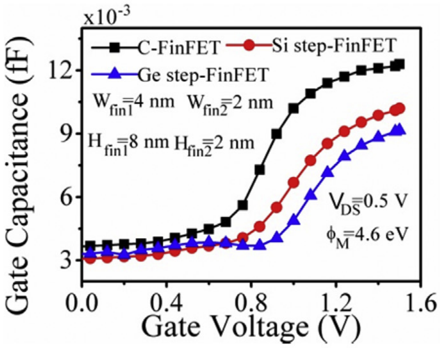
Orientation	Probability	Work function
<200>	60%	4.6 eV
<111>	40%	4.4 eV



(a)



(b)



(c)

Fig. 2. Comparison of (a) Drain current (μA) of proposed FinFETs with C- FinFET in log scale, (b) Drain current (μA) of proposed FinFETs with C- FinFET in linear scale, and (c) Gate Capacitance (fF) of proposed FinFETs with C-FinFET.

Gate capacitance (C_{gg}) decides the intrinsic delay of any device. Fig. 2(c) depicts the comparison of C_{gg} among C-FinFET and step-FinFETs with same dimensions. As fin width reduces, the side wall area under the gate reduces leading to a reduction in gate capacitance of step-FinFETs.

C_{gg} of Ge step-FinFET is low at high gate voltage and high at low gate voltage than Si step-FinFET as portrayed in Fig. 2(c). C_{gg} value depends on oxide capacitance (C_{ox}), depletion capacitance (C_b), and inversion capacitance (C_i). At low value of gate voltage, only depletion capacitance will be there, which is directly proportional to square root dielectric constant of substrate. Hence at low gate voltage C_{gg} value of Ge is large than Si. However, due to narrower band gap of Ge, C_{gg} at high gate voltage falls down as compared to its Si counterpart.

3.2. Parametric analysis of Si step-FinFET and Ge step-FinFET

A parametric analysis with $H_{fin1} = 8$ nm, $H_{fin2} = 2$ nm, $W_{fin1} = 4$ nm, and $W_{fin2} = 2$ nm is presented for Si and Ge step-FinFETs. The effect of channel length and oxide thickness, on drain current, gate capacitance (C_{gg}), intrinsic delay, I_{on}/I_{off}, SS, DIBL, and V_T is discussed. The WF of gate polysilicon and drain-to-source voltage (V_{DS}) are considered as 4.6 eV and 0.5 V, respectively.

Fig. 3(a) shows the drain current vs. gate voltage for $L_g = 8$ nm and using oxide thickness as a parameter. As expected, a reduction of oxide thickness increases the gate control over the channel, and hence, the on current increases for both the devices. But Ge device shows higher on current and lower off current than Si device due to lower band gap. Fig. 3(b) shows the drain current vs. gate voltage using L_g as a parameter. It is observed that as the channel length increases, the drain current decreases for both saturation and linear regions in both the devices. This is because an increased channel length lowers the channel potential resulting in a decreased drain current. However, for every channel length Ge step-FinFET shows better on current as well as off current. Thus, the Ge step-FinFET can tolerate more scaling effects so far as current characteristic is concerned.

A parametric analysis of C_{gg} vs. gate voltage for different oxide thickness and $L_g = 8$ nm is portrayed in Fig. 3(c). As oxide thickness is reduced, gate control over channel increases which leads to an increase in C_{gg} for both the devices. C_{gg} of Ge step-FinFET is lower at higher gate voltage and higher at lower gate voltage. The reason for the same is as explained for Fig. 2(b). A scaled down device has a lower gate area, and hence, gives rise to a lower C_{gg} as shown in Fig. 3(d).

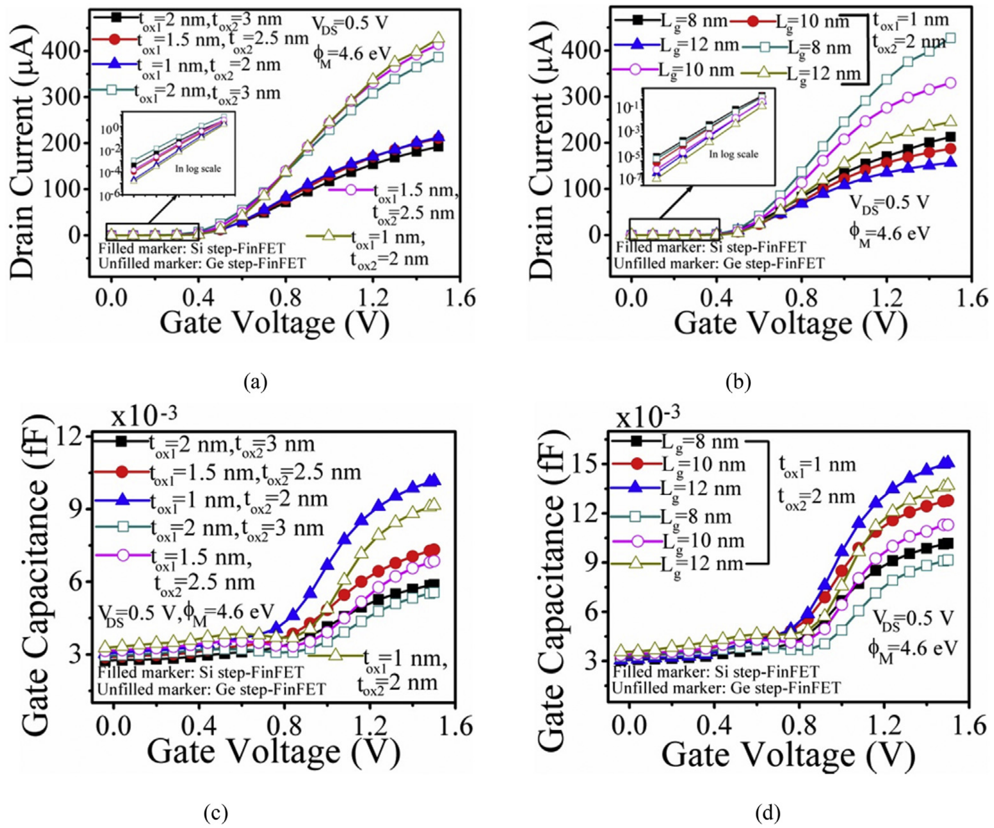


Fig. 3. Comparison between Si step-FinFET and Ge step-FinFET (a) Drain current (μA) vs. gate voltage (V) for various oxide thickness at $L_g = 8$ nm, (b) Drain current (μA) vs. gate voltage (V) for various channel length at $t_{ox1} = 1$ nm and $t_{ox2} = 2$ nm, (c) Gate Capacitance (fF) vs. gate voltage (V) for various oxide thickness at $L_g = 8$ nm, and (d) Gate Capacitance (fF) vs. gate voltage (V) for various channel length at $t_{ox1} = 1$ nm and $t_{ox2} = 2$ nm.

The intrinsic delay defines the switching speed of any circuit. To have higher switching speed, a lower intrinsic delay is necessary. Intrinsic delay ($C_{gg}V_{DS}/I_{on}$) vs. oxide thickness is plotted in Fig. 4(a). A lower oxide thickness gives rise to a higher C_{gg}, which in turn, increases the intrinsic delay. But Ge device has lesser C_{gg} and higher I_{on}; therefore, with oxide thickness scaling Ge step-FinFET poses lesser intrinsic delay than Si step-FinFET. A parametric analysis of intrinsic delay vs. gate voltage at various L_g is portrayed in Fig. 4(b). As the gate length is reduced both the operations: speed and number of components on a chip increases. On the other hand, when we scale L_g we get a lower C_{gg} and higher on current as shown in Fig. 3(d) and (b), respectively. This explains a lower intrinsic delay with scaled L_g in general. But, due to lower C_{gg} and higher I_{on}, Ge step-FinFET

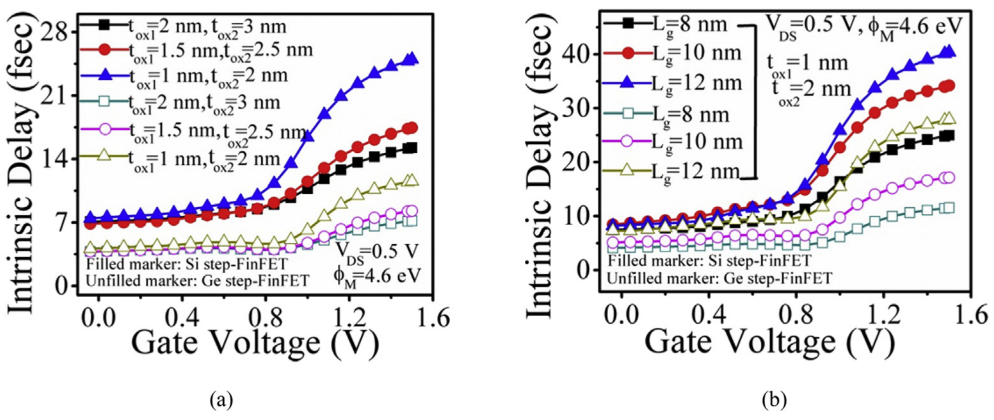


Fig. 4. Comparison intrinsic delay between Si step-FinFET and Ge step-FinFET (a) Intrinsic delay (fsec) vs. gate voltage (V) for various oxide thickness at $L_g = 8$ nm, and (b) Intrinsic delay (fsec) vs. gate voltage (V) for various channel length at $t_{ox1} = 1$ nm and $t_{ox2} = 2$ nm.

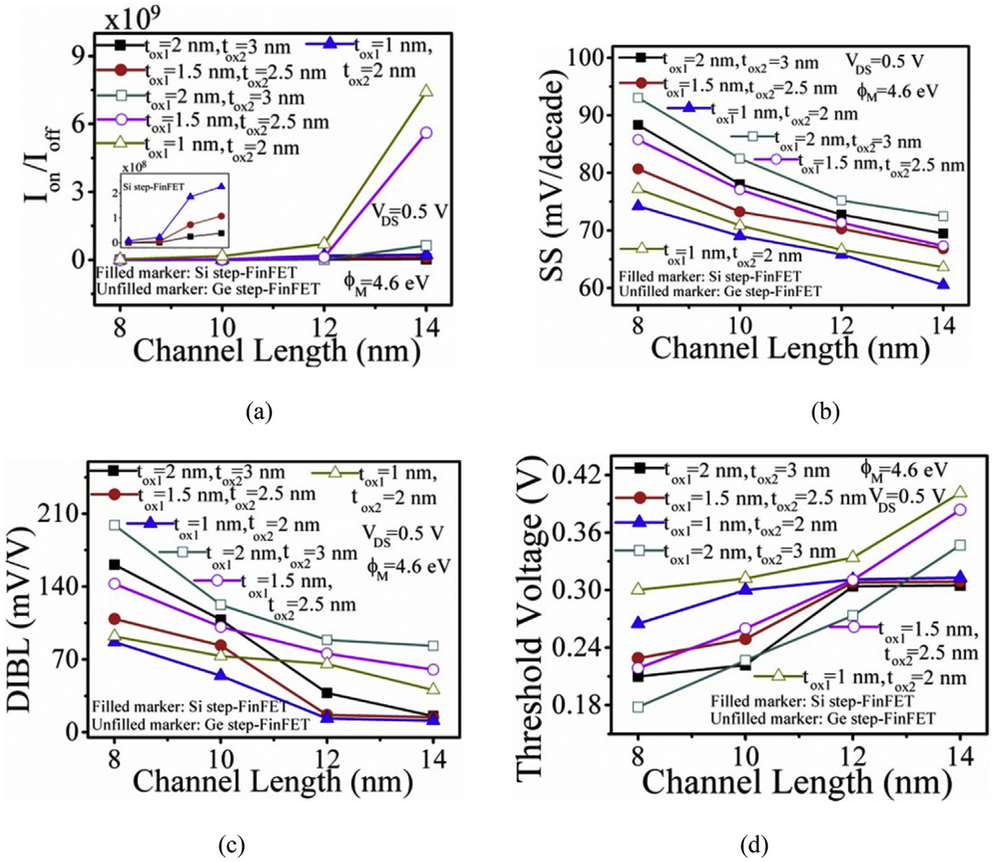


Fig. 5. Comparison of SCEs between Si step-FinFET and Ge step-FinFET (a) Current ratio vs. channel length (nm), (b) SS (mV/decade) vs. channel length (nm), (c) DIBL (mV/V) vs. channel length (nm), and (d) V_t roll off (V) vs. channel length (nm).

has lower intrinsic delay. Intrinsic delays of 15.2 fsec and 7.18 fsec, have been obtained for the structural combination of $L_g = 8$ nm, $t_{ox1} = 2$ nm and $t_{ox2} = 3$ nm at gate voltage 1.5 V for Si and Ge step-FinFETs, respectively.

Fig. 5(a) shows the variation of current ratio vs. channel length for both the devices. As the oxide thickness reduces, gate control over the channel increases and results in increased I_{on}/I_{off} ratio. With increase in channel length, off current reduces more significantly than on current which leads to improvement in I_{on}/I_{off}. As Ge device shows higher on and lesser off current, it provides higher I_{on}/I_{off}, irrespective to the thickness.

Variations of subthreshold swing (SS) as a function of channel length is plotted in Fig. 5(b). We know that the standard expression for SS is given by Ref. [34]:

$$SS = 2.3 \ \eta \ V_T \tag{1}$$

where V_T is the thermal voltage and η, called subthreshold slope, and is given by Ref. [34]:

$$\eta = 1 + \frac{C_{dep}}{C_{ox}} \tag{2}$$

where C_{dep} and C_{ox} are depletion and oxide capacitance, respectively.

As the channel length is reduced, the source to drain distance is reduced, resulting in reduction of effective depletion width causing an increase in C_{dep}. This explains why we get an enhanced SS with a reduction in channel length for both the devices. However, at a fixed gate length, as oxide thickness is reduced, C_{ox} increases which leads to a lowering of SS value. But due to higher dielectric constant, Ge step-FinFET has higher SS value as seen in the plots. In the case of Si step-FinFET, an SS value of around 60.54 mV/dec is observed for $L_g = 14$ nm, $t_{ox1} = 1$ nm, and $t_{ox2} = 2$ nm.

From Fig. 5(c), it is observed that DIBL value increases with the decrease in channel length at fixed oxide thickness for both the devices. As the channel length is scaled down, the channel control is more shared by the drain which enhances the effects of lateral field resulting in an increased DIBL effect. However, with the reduction of t_{ox}, oxide capacitance increases and the control on channel by the gate increases leading to a reduction in the same. Due to higher dielectric constant, Ge step-FinFET

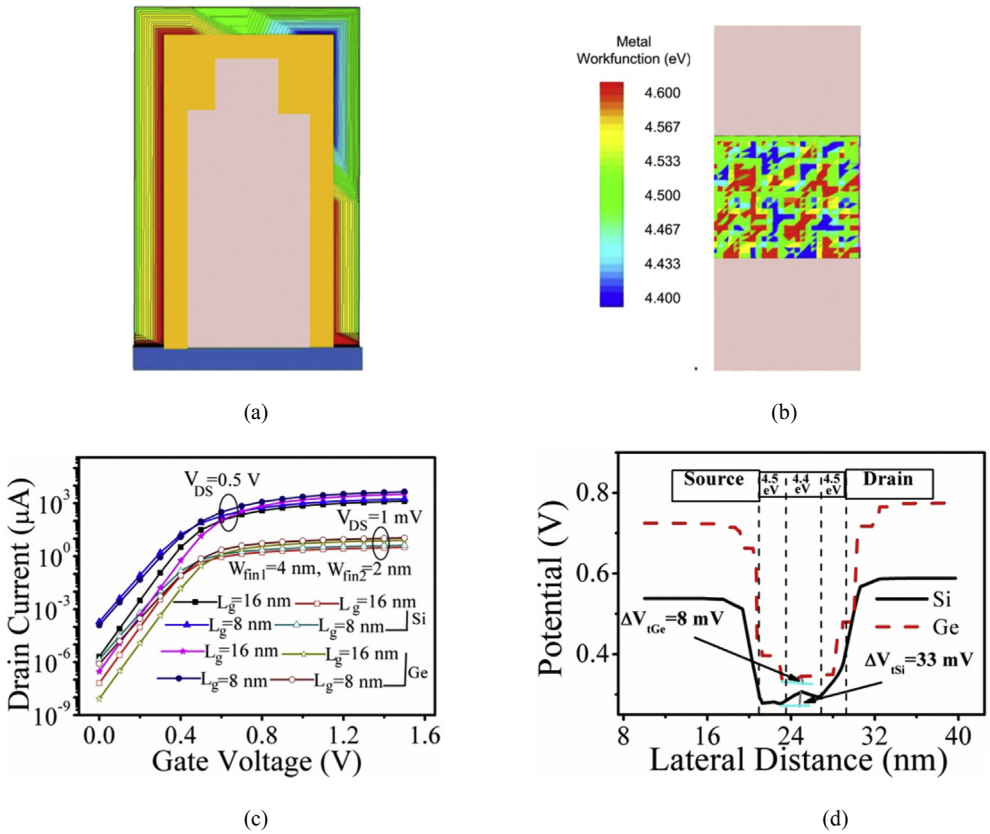


Fig. 6. (a) Front view of simulated step-FinFET, (b) Top view of simulated step-FinFET showing local variation of metal-gate work function, (c) Drain Current (μA) vs. Gate Voltage (V) for a particular grain pattern both for Ge and Si step-FinFET at two different V_{DS}, and (d) Potential within the channel of the proposed device at V_{GS}=V_T and V_{DS} = 50 mV for both Si and Ge materials for the combination L_g = 8 nm, t_{ox1} = 1 nm and t_{ox2} = 2 nm.

has more DIBL value than Si step-FinFET. DIBL = 11.11 mV/V is observed for the combination of L_g = 14 nm, t_{ox1} = 1 nm, and t_{ox2} = 2 nm of Si step-FinFET.

Fig. 5(d) shows the comparison of threshold voltage vs. channel length for various oxide thicknesses. Threshold voltage is extracted by constant current method (=10^{-7} A). As seen, that the V_T decreases with decrease in channel length. However, the enhanced gate control with reduced oxide thickness increases the V_T. It changes at a faster rate for lower channel length for both the devices. But, the V_T roll off is more for Ge device than Si device due to higher dielectric constant of Ge.

3.3. Comparison of work function variation of conventional FinFET, Si step-FinFET, and Ge step-FinFET

Fig. 6(a) and (b) show the simulated front and top views of step-FinFET with local variation of V_T. As Ti metal is randomly distributed, we have simulated each device 200 times for each combination of structural parameters, where each device simulation has unique grain pattern. To study the effect of WFV on performance parameters of both Si and Ge step-FinFETs, the variability of threshold voltage (σV_T), subthreshold swing (σSS), on current (σI_{on}), off current (σI_{off}), and current ratio ($\sigma(I_{on}/I_{off})$) are extracted from the transfer characteristics. The standard deviations of different parameters are listed in Tables 3 and 4, corresponding to V_{DS} = 0.5 V. Constant current method (10^{-7} A) is adopted to extract the V_T.

Table 3
Comparison of standard deviation of V_T and SS between C-FinFET and Si step-FinFET.

SET	Combination of structure parameters	C-FinFET		Si step-FinFET	
		σV_T (mV)	σSS (mV/dec)	σV_T (mV)	σSS (mV/dec)
1	L_g = 16 nm,W_{fin1} = 4 nm,W_{fin2} = 2 nm, t_{ox1} = 1 nm,t_{ox2} = 2 nm	40.15	2.48	39.14	0.989
2	L_g = 8 nm,W_{fin1} = 4 nm,W_{fin2} = 2 nm, t_{ox1} = 1 nm,t_{ox2} = 2 nm	45.92	2.45	44.11	0.87

Table 4
Comparison of standard deviation of I_{on}, I_{off} and I_{on}/I_{off} between C-FinFET and Si step-FinFET.

SET	Combination of structure parameter	C-FinFET			Si step-FinFET		
		σI_{on} (μA)	σI_{off} (nA)	$\sigma(I_{on}/I_{off})$	σI_{on} (μA)	σI_{off} (nA)	$\sigma(I_{on}/I_{off})$
1	$L_g = 16$ nm,$W_{fin1} = 4$ nm,$W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	24.7	0.489	1.43×10^7	61.8	0.086	9.92×10^8
2	$L_g = 8$ nm,$W_{fin1} = 4$ nm,$W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	32	1.07	1.99×10^6	34.8	1.22	1.59×10^7

Table 5
Comparison of standard deviation of V_T and SS between Si step-FinFET and Ge step-FinFET.

SET	Combination of structure parameter	Ge		Si	
		σV_T (mV)	σSS (mV/dec)	σV_T (mV)	σSS (mV/dec)
1	$L_g = 16$ nm,$W_{fin1} = 4$ nm,$W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	33.05	1.14	39.14	0.989
2	$L_g = 8$ nm,$W_{fin1} = 4$ nm,$W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	40.76	0.976	44.11	0.87
3	$L_g = 8$ nm,$W_{fin1} = 2$ nm,$W_{fin2} = 1$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	51.82	1.05	59.27	0.91
4	$L_g = 8$ nm,$W_{fin1} = 4$ nm,$W_{fin2} = 2$ nm, $t_{ox1} = 0.8$ nm, $t_{ox2} = 1.8$ nm	38.83	0.901	42.46	0.81

3.3.1. *Comparison of WFV between conventional FinFET and Si step-FinFET*

Table 3 gives the comparison of variation of V_T and SS between C-FinFET and Si step-FinFET. Comparison of standard deviation of I_{on}, I_{off}, and I_{on}/I_{off} between C-FinFET and Si step-FinFET is reported in Table 4. From Table 3, it is observed that the variations in both the V_T and SS are higher for C-FinFET than Si step-FinFET. σV_T increases with reduction of the channel length for both the devices. Moreover, from Table 4, we observe that for each structural combination, the variation in I_{on} is less than that of I_{off} in C-FinFET. This behaviour is due to lower off current and higher on current of Si step-FinFET than C-FinFET. But $\sigma(I_{on}/I_{off})$ of Si step-FinFET is larger than C-FinFET.

3.3.2. *Comparison of WFV between Si step-FinFET and Ge step-FinFET*

It is observed from Table 5 that higher variation in V_T occurred for Si step-FinFET compared to Ge step-FinFET. The better performance of Ge device, in this regard, is again due to higher dielectric constant. It is also seen that when channel length reduced from 16 nm to 8 nm, σV_T increases at a faster rate both the devices. The origin of such behaviour stems from the fact that a decrease in L_g reduces the gate area leading to a decrease in the number grains. Similarly, as the fin width is reduced the effective gate area also decreases and the fluctuation in V_T increases of as shown in rows 2 and 3 of Table 5. These behaviours are consistent with the existing literature [18]. It is also perceptible that minimal variation of V_T occurs with the scaling of oxide thickness as reported in rows 2 and 4.

It is seen from Table 5 that the variation in SS due to WFV of Ge is higher compared to Si, irrespective to the combinations of structural parameters. Hence, Ge step-FinFET is less resistant to SCE than the Si step-FinFET. SCE is directly related with dielectric constant of the channel material; with increase in dielectric constant of channel material SCE effect also increases. This is the reason why the Ge step-FinFET is less immune to SCE [35,36]. Lower variation in SS for Si device is attributed to better SCE. The variation in SS decreases with the scaling of L_g, keeping other parameters fixed for both the devices. Likewise, a reduction of fin width increases the variation of SS in both the Ge and Si step-FinFETs. On the other hand, SS decreases with the reduction in oxide thickness. Such variations in SS may be attributed to increase or decrease in electrostatic integrity [37] of the device.

From Table 6, it is observed that variation in on current and off current, respectively, are higher and lower in Ge step-FinFET compared to Si step-FinFET for all structural combinations. But the variation of I_{on}/I_{off} is more of Ge step-FinFET. This behaviour of Ge is due to narrower band gap over Si, which is responsible for higher drive current and lower value off current. With the reduction of oxide thickness, the drive current increases and variation of I_{on}/I_{off} ratio increases as shown in rows 2 and 4 of Table 6. When the fin width is reduced, the fluctuations of on and off current decrease both for Ge and Si step-FinFETs, but $\sigma(I_{on}/I_{off})$ increases(rows 2 and 3). This is due to the fact that a decrease in fin width increases the source/drain resistances which results in a decrease of current. With the scaling of L_g since I_{off} increases variation of $\sigma(I_{on}/I_{off})$ reduces.

Table 6
Comparison of standard deviation of I_{on}, I_{off}, and I_{on}/I_{off} between Si step-FinFET and Ge step-FinFET.

SET	Combination of structure parameter	Ge			Si		
		σI_{on} (μA)	σI_{off} (nA)	$\sigma(I_{on}/I_{off})$	σI_{on} (μA)	σI_{off} (nA)	$\sigma(I_{on}/I_{off})$
1	$L_g = 16$ nm,$W_{fin1} = 4$ nm, $W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	108	0.0028	2.12×10^{10}	61.8	0.086	9.92×10^8
2	$L_g = 8$ nm,$W_{fin1} = 4$ nm, $W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	109	0.0713	3.98×10^7	34.8	1.22	1.59×10^7
3	$L_g = 8$ nm,$W_{fin1} = 2$ nm, $W_{fin2} = 1$ nm, $t_{ox1} = 1$ nm,$t_{ox2} = 2$ nm	108	0.0292	7.05×10^9	31	0.101	7.19×10^8
4	$L_g = 8$ nm,$W_{fin1} = 4$ nm, $W_{fin2} = 2$ nm, $t_{ox1} = 0.8$ nm, $t_{ox2} = 1.8$ nm	243	0.261	1.12×10^8	38.2	0.688	2.13×10^7

Table 7
Comparison of standard deviation of V_T and SS due to scaling of V_{DS} for the combination $L_g = 8$ nm, $W_{fin1} = 2$ nm, $W_{fin2} = 1$ nm, $t_{ox1} = 1$ nm, and $t_{ox2} = 2$ nm.

V_{DS}	Ge		Si	
	σV_T (mV)	σSS (mV/dec)	σV_T (mV)	σSS (mV/dec)
0.5	51.82	1.05	59.27	0.91
0.4	56.68	1.16	60.59	1.12

Table 7 gives the fluctuation of V_T and SS with the scaling of V_{DS}, which shows that as V_{DS} is scaled, we get an increased in σV_T as well as σSS. However, the Si device gives more variation in V_T, while the Ge device gives more fluctuation in SS.

The electrical characteristics at $H_{fin1} = 8$ nm and $H_{fin2} = 2$ nm for a particular grain pattern at low and high values of V_{DS} are as shown in Fig. 6(c). For same channel length, Ge step-FinFET offers lower off current and higher on current. This behaviour is due to narrower band gap and high mobility in Ge. Furthermore, at $V_{DS} = 1$ mV drain current for both Ge and Si decreases.

Fluctuation in channel potential due to variation in work function of the channel depends on the dielectric constant of the material. To verify this statement, we divide the channel into three regions. The two regions: near the source and drain having equal work functions (4.5 eV) and the third region: middle of channel has WF equal to 4.4 eV. The drain to source voltage is equal to 50 mV and gate voltage is equal to the threshold voltage. As evidenced in Fig. 6(d), fluctuation in the magnitude of the channel potential is more for Si step-FinFET for same value of metal gate work function. Such a higher fluctuation gives a higher variation of the threshold voltage.

In Fig. 7(a), σV_T vs. inverse of square root of effective area is shown. Such plot, which defines the matching of proposed device due to variation in length and width of the device, is known as Pelgrom's plot [38]. A proportional parameter A_{vt}, which is defined as the slope of curves, is a measure of the threshold voltage variability. As higher value of A_{vt} is observed for Si step-FinFET, it has more variation in threshold voltage.

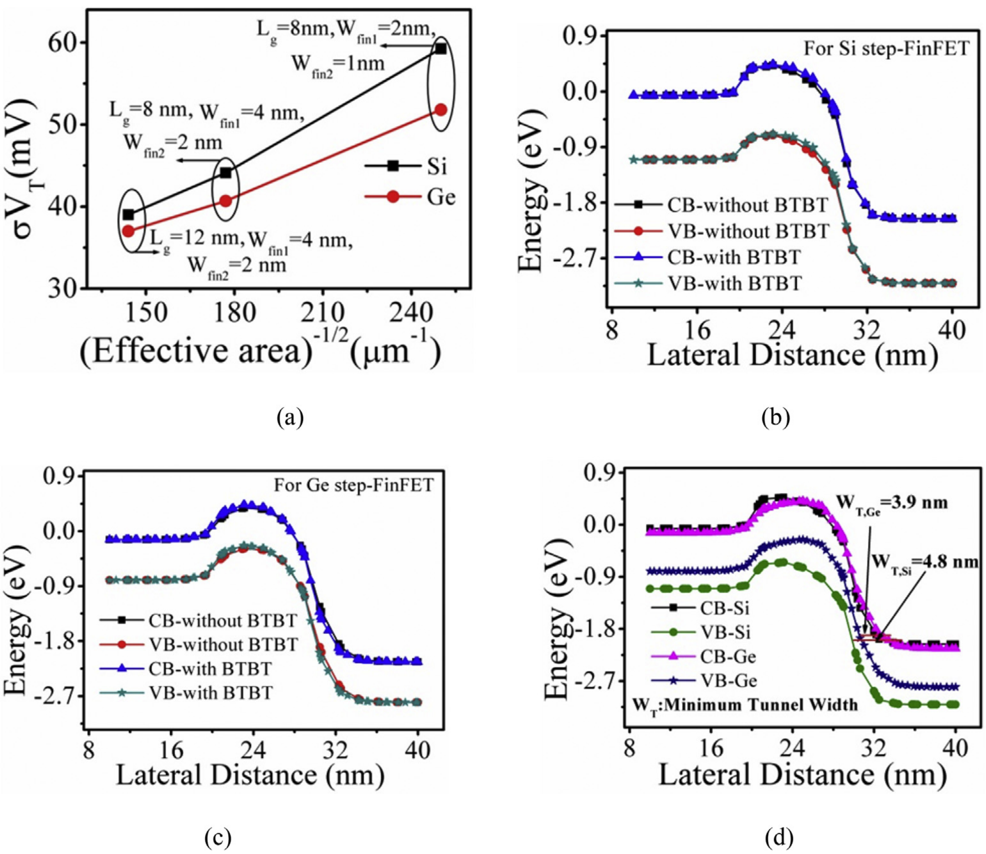


Fig. 7. (a) Pelgrom's plot for the Si and Ge step-FinFET, (b) Band Diagram at off-state with and without BTBT for $L_g = 8$ nm, $W_{fin1} = 4$ nm, $W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm, $t_{ox2} = 2$ nm and $V_{DS} = 0.5$ V of Si step-FinFET, (c) Band Diagram at off-state with and without BTBT for $L_g = 8$ nm, $W_{fin1} = 4$ nm, $W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm, $t_{ox2} = 2$ nm and $V_{DS} = 0.5$ V of Ge step-FinFET, and (d) Comparison of band diagram at off state between the Si and Ge step-FinFET to show the minimum tunnel width.

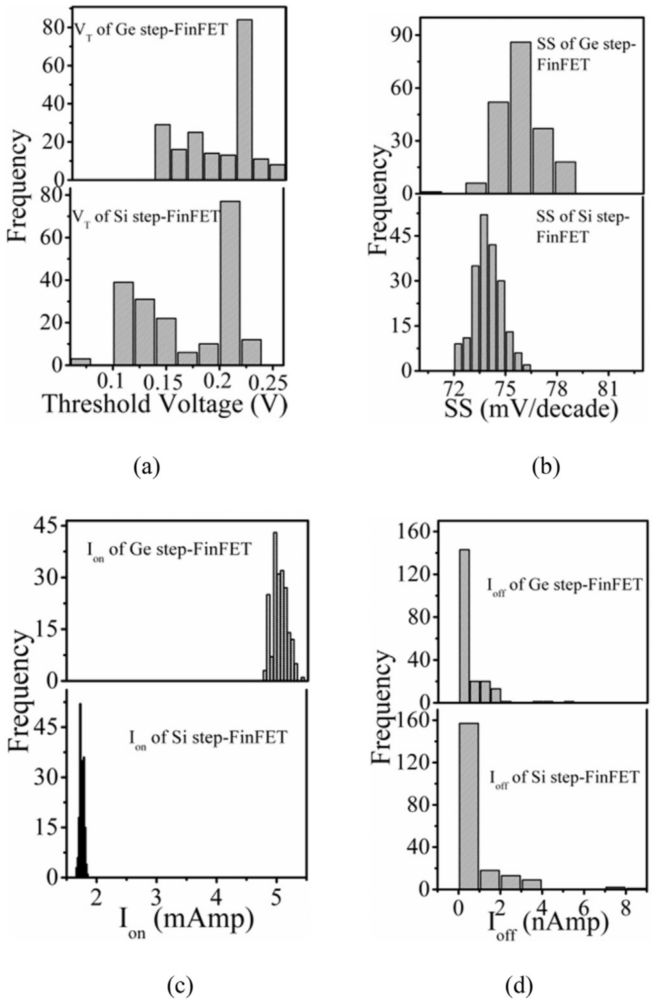


Fig. 8. Histograms due to WFV for Ge and Si step-FinFET for $L_g = 8$ nm, $W_{fin1} = 4$ nm, $W_{fin2} = 2$ nm, $t_{ox1} = 1$ nm, $t_{ox2} = 2$ nm, and $V_{DS} = 0.5$ V (a) V_T fluctuation, (b) SS fluctuations, (c) I_{on} fluctuations, and (d) I_{off} fluctuations.

As the device is in nanoscale, the band to band tunnelling (BTBT) may also affect the results. In order to realize the effect of BTBT, the energy band diagram at off state with and without considering BTBT model is shown in Fig. 7(b) and (c). Also we have activated Hurkx BTBT model [32] in our simulation. It is observed from Fig. 7(b) and (c) that the energies of conduction band (CB) and valence band (VB) coincide with and without the use of BTBT model for both the devices. Therefore, our proposed device is free from BTBT.

We have also investigated of the off state tunnel width of the devices. It is observed from Fig. 7(d) that the tunnel width is more for Si step-FinFET compared to Ge step-FinFET. The larger band gap produces larger tunnel width resulting in no BTBT in Si device.

The frequency distribution of V_T and SS correspond to $V_{DS} = 0.5$ V for the combination of structure in row 2 (Table 5) is shown Fig. 8. The nominal value of V_T variation is more for Si step-FinFET and SS variation is high for Ge step-FinFET, which is consistent with existing literature [39]. This observation is also a part of the evidences that Si is more immune to SCEs. It is also seen that the distribution of V_T and SS are neither Gaussian nor symmetric.

Fig. 8(c) and (d) show the histograms of I_{on} and I_{off} for Ge and Si step-FinFET, respectively. It is observed that the variation of I_{on} is more for Ge step-FinFET compared to Si step-FinFET and I_{off} variation is less for Ge step-FinFET. Therefore, these variations again attributed that the variation of I_{on}/I_{off} ratio is more for Ge step-FinFET than Si step-FinFET. The distribution of I_{on} and I_{off} are also neither Gaussian nor symmetric.

4. Conclusion

A comparative simulation study among the proposed devices, Si and Ge based step-FinFET and C-FinFET is reported. Results show that the Ge step-FinFET has better on current, lower off current, lower value of gate delay, and better I_{on}/I_{off} ratio.

On the other hand, parametric analysis reveals that the Si step-FinFET offers impressive control on SCEs with the scaling of oxide thicknesses and channel length than Ge step-FinFET. However, Ge step-FinFET gives better I_{on}/I_{off} ratio, lower intrinsic delay than Si step-FinFET for a given oxide thickness and channel length.

Furthermore, the effect of the work function variability (WFV) of Ti metal gate shows that the variabilities of threshold voltage (σV_T) and subthreshold swing (σSS) of C-FinFET are higher than Si step-FinFET. But the variability of current ratio ($\sigma I_{on}/I_{off}$) is less in the case of C-FinFET. In a consecutive step, a comparative study between Si and Ge step-FinFETs is made in presence of WFV. It is observed that Ge step-FinFET is more immune to WFV in case of consideration of threshold voltage variability (σV_T). However, the variabilities of subthreshold swing (σSS) and current ratio ($\sigma I_{on}/I_{off}$)) are more for Ge step-FinFET than Si step-FinFET. Thus, the WFV has a significant effect on electrical properties of FinFET, which can be reduced with suitable device geometry. The proposed device can perform better in low power applications.

References

[1] K. Roy, S. Mukhopadhyay, H. M Meimand, Leakage current mechanisms and leakage reduction techniques in deep-submicrometer CMOS circuits, Proc. IEEE 91 (2) (April 2003) 305–327, http://dx.doi.org/10.1109/JPROC.2002.808156.
[2] D.J. Frank, R.H.E. Nowak, P.M. Solomon, Y. Taur, H.S.P. Wong, Device scaling limits of Si MOSFETs and their application dependencies, Proc. IEEE 89 (3) (2001) 259–288, http://dx.doi.org/10.1109/5.915374.
[3] H.-S.P. Wong, Beyond the conventional transistor, IBM J. Res. Dev. 46 (2/3) (2002) 133–168.
[4] Y.C. Yeo, T.J. King, C. Hu, MOSFET gate leakage modeling and selection guide for alternative gate dielectrics based on leakage considerations, IEEE Transac. Electron Devices 50 (4) (2003) 1027–1035, http://dx.doi.org/10.1109/TED.2003.812504.
[5] J. Chen, T.Y. Chan, I.C. Chen, P.K. Ko, C. Hu, Subbreak- down drain leakage current in MOSFET, IEEE Electron device Lett. 8 (11) (1987) 515–517, http://dx.doi.org/10.1109/EDL.1987.26713.
[6] G.E. Moore, Cramming more components onto integrated circuits, Electronics 38 (8) (1965) doi: 10.0018-9219(98)00753-1.
[7] D. Bhattacharya, N.K. Jha, FinFETs: from Devices to Architectures, Article ID 365689, Hindawi Publishing Corporation, Advances in Electronics, 2014, p. 21. doi: 10.1155/2014/365689.
[8] N. Fasarakis, A. Tsormpatzoglou, et al., Compact model of drain current in short-channel triple-gate FinFETs, IEEE Transac. Electron Devices 59 (7) (2012) 1891–1898, http://dx.doi.org/10.1109/TED.2012.2195318.
[9] S.W.M. Hatta, N. Soin, S.H.A. Rahman, Y.A. Wahab, H. Hussain, Effects of fin width variation on the performance of 16nm FinFETs with round fin corners and tapered fin shape, Kuala Lumpur, Proc. IEEE-ICSE (2014) 533–536, http://dx.doi.org/10.1109/SMELEC.2014.6920916.
[10] S. Raj, J. Sahu, W. Dattatray, R.A. Mishra, S. Tiwari, Modelling,Design and Performance Comparison of Triple Gate Cylindrical and Pertially Cylindrical FinFETs for Low-power Applications, ISRN Electronics, 2012, http://dx.doi.org/10.5402/2012/827452.
[11] V. Narendar, R.A. Mishra, Analytical modeling and simulation of multigate FinFET devices and the impact of high k dielectrics on short channel effects (SCEs), Superlattices Microstruct 85 (2015) 357–369, http://dx.doi.org/10.1016/j.spmi.2015.06.004.
[12] C.W. Sohn, C.Y. Kang, et al., Effect of Fin Height of Tapered FinFETs on the Sub-22-nm System on Chip (SoC) Application Using TCAD Simulation, Int. Symp. on VLSI Technology, Systems, and Applications (VLSI-TSA), 2013, http://dx.doi.org/10.1109/VLSI-TSA.2013.6545634.
[13] R. Ritzenthaler, F. Lime, B. Iñiguez, O. Faynot, S. Cristoloveanu, 3D analytical modelling of subthreshold characteristics in Pi-gate FinFET transistors, Proc. Eur. Solid-State Device Res. Conf. (ESSDERC) (2010), http://dx.doi.org/10.1109/ESSDERC.2010.5618179.
[14] M.S. Yeh, Y.C. Wu, et al., Fabrication, characterization and simulation of Ω-gate twin poly-Si FinFET nonvolatile memory, Nanoscale Res. Lett. 8 (1) (2013) 1–5, http://dx.doi.org/10.1186/1556-276X-8-331.
[15] M. Mehrad, A.A. Orouji, Partially cylindrical fin field-effect transistor: a novel device for nanoscale applications, IEEE Transac. Device Mater. Reliab. 10 (2010) 271–275, http://dx.doi.org/10.1109/TDMR.2010.2046663.
[16] R. Das, R. Goswami, S. Baishya, Tri-gate heterojunction SOI Ge-FinFETs, Superlattices Microstruct 91 (2016) 51–61, http://dx.doi.org/10.1016/j.spmi.2015.12.039.
[17] A. Cottrel, Introduction to the Modern Theory of Metals, Inst. Metals, London, U.K., 1988.
[18] H.F. Dadgour, K. Endo, V.K. De, K. Banerjee, Grain-orientation induced work function variation in nanoscale metal-gate Transistors—Part I: modeling, analysis, and experimental validation, IEEE Transac. Electron Devices 57 (10) (2010) 2504–2514, http://dx.doi.org/10.1109/TED.2010.2063191.
[19] A.R. Brown, N.M. Idris, J.R. Watling, A. Asenov, Impact of metal gate granularity on threshold voltage variability: a full-scale three-dimensional statistical simulation study, IEEE Electron Device Lett. 31 (11) (2010) 1199–1201, http://dx.doi.org/10.1109/LED.2010.2069080.
[20] Y. Li, H.W. Cheng, C.Y. Yiu, H.W. Su, Nanosized metal grains induced electrical characteristic fluctuation in 16-nm-gate high-k/metal gate bulk FinFET devices, Microelectron. Eng. 88 (2011) 1240–1242, http://dx.doi.org/10.1016/j.mee.2011.03.037.
[21] T. Matsukawa, S. Ouchi, K. Endo, Comprehensive analysis of variability sources of FinFET characteristics, Proc. Symp. VLSI Technol. (Jun. 2009) 118–119.
[22] S.M. Nawaz, S. Dutta, A. Chattopadhyay, A. Mallik, Comparison of random dopant and gate-metal workfunction variability between junctionless and conventional FinFETs, IEEE Electron Device Lett. 35 (6) (2014) 663–665, http://dx.doi.org/10.1109/LED.2014.2313916.
[23] S.M. Nawaz, A. Dutta, A. Mallik, Comparison of gate-metal work function variability between Ge and Si p-Channel FinFETs, IEEE Transac. Electron Devices 62 (12) (2015) 3951–3956, http://dx.doi.org/10.1109/TED.2015.2488101.
[24] R. Saha, B. Bhowmick, S. Baishya, Statistical dependence of gate metal work function on various electrical parameters for an n-channel Si step-FinFET, IEEE Trans. Electron Devices 64 (3) (March 2017) 969–976, http://dx.doi.org/10.1109/TED.2017.2657233.
[25] T.P. Ma, Making silicon nitride film a viable gate dielectric, Trans. Electron Devices 45 (3) (1998) 680–690 doi: S 0018-9383(98)01670-0.
[26] C.C. Hobbs, L.R.C. Fonseca, et al., Fermi-level pinning at the polysilicon/metal oxide interface: Part I, IEEE Transac. Electron Devices 51 (6) (2004) 971–977, http://dx.doi.org/10.1109/TED.2004.829510.
[27] E.P. Gusev, V. Narayanan, M.M. Frank, Advanced high-k dielectric stacks with poly-Si and metal gates: recent progress and cur-rent challenges, IBM J. Res. Dev. 50 (4/5) (2006) 387–410, http://dx.doi.org/10.1147/rd.504.0387.
[28] C. Le. Royer, L. Clavelier, C. Tabone, R. Truche, 105 nm gate length p MOSFETs with high-K and metal gate fabricated in a Si process line on 200 mm GeOI wafers, Solid-State Electron. 52 (9) (2008) 1285–1290, http://dx.doi.org/10.1016/j.sse.2008.04.019.
[29] C. Le, Royer, Interfaces and performance: what future for nanoscale Ge and SiGe based CMOS, Microelectron. Eng. 88 (7) (2011) 1541–1548, http://dx.doi.org/10.1016/j.mee.2011.03.052.
[30] Y. Liu, S. Kijima, M. Masuhara, et al., Investigation of the TiN gate electrode with tunable work function and its application for FinFET fabrication, IEEE Transac. Nanotechnol. 5 (6) (2006) 723–728, http://dx.doi.org/10.1109/TNANO.2006.885035.
[31] H. Dadgour, K. Endo, V. De, K. Banerjee, Modeling and analysis of grain-orientation effects in emerging metal gate devices and implications for SRAM reliability, IEDM Tech. Dig. (2008) 705–708, http://dx.doi.org/10.1109/IEDM.2008.4796792.
[32] Synopsys, Sentaurus Device User, 2012.
[33] R. Rios, et al., Comparison of junctionless and conventional trigate transistors with L_g down to 26 nm, IEEE Electron Device Lett. 32 (9) (Sept. 2011) 1170–1172, http://dx.doi.org/10.1109/LED.2011.2158978.
[34] Y. Tsividis, C. McAndrew, Operation and Modelling of the MOS Transistor, second ed., Oxford University, 2011.
[35] A. Tsormpatzoglou, C.A. Dimitriadis, et al., Semi-analytical modeling of short-channel effects in Si and Ge symmetrical double-gate MOSFETs, IEEE Transac. Electron Devices 54 (8) (2007) 1943–1952, http://dx.doi.org/10.1109/TED.2007.901075.

[36] K. Krishnamohan, D. Kim, C.D. Nguyen, C. Jungemann, Y. Nishi, K.C. Saraswat, High-mobility low band-to-band-tunneling strained-germanium double-gate heterostructure FETs: simulations, IEEE Transac. Electron Devices 53 (5) (2006) 1000–1009, http://dx.doi.org/10.1109/TED.2006.872367.
[37] R. Wang, T. Yu, R. Huang, Y. Wang, Impacts of short-channel effects on the random threshold voltage variation in nanoscale transistors, Sci. China Inf. Sci. 56 (6) (2013), http://dx.doi.org/10.1007/s11432-013-4814-9, 062403–1–062403-7.
[38] M.J.M. Pelgrom, A.C.J. Duinmaijer, A.G. Welbers, Matching properties of MOS transistors, IEEE J. Solid-state circuit 24 (5) (1989) 1433–1440.
[39] Y.-S. Wu, C.-H. Chiang, P. Su, Investigation of scalability for Ge and InGaAs channel multi-gate NMOSFETs, Procc. Int. Symp.VLSI-TSA (April 2012) 1–2, http://dx.doi.org/10.1109/VLSI-TSA.2012.6210132.